Test Evidence

Failure-case tests, experiments, and what the results really say

CoolRoute: Pedestrian Routing with Shade and Heat-Exposure Budgets

The hackathon asks us to **test failure cases** and **show evidence**. This file lists every check we ran and what we found, including results that are not flattering.

1. Summary

Check	Result
Routing engine tests (Python)	13 of 13 passed
Backend API tests over HTTP	12 of 12 passed
JavaScript engine versus Python engine	60 of 60 random scenarios gave identical routes and heat
Large experiment	148 random long trips at 1 pm with a 60% budget

2. Engine tests (failure cases first)

#	Test	Result	Detail
1	Sun is zero at midnight	PASS	
2	Sun peaks near noon	PASS	
3	Cool route heat <= budget	PASS	heat 455 <= 457
4	Cool route is cooler than baseline	PASS	
5	Baseline is the shortest	PASS	
6	At night heat is zero and routes match	PASS	
7	Without shade the baseline heat is higher	PASS	
8	Closed footpath is never used	PASS	
9	Step-free route uses only step-free edges	PASS	
10	No step-free route -> error returned	PASS	
11	Impossible budget is flagged (budget_met False)	PASS	
12	Missing-data edges exist in the map and are counted	PASS	9 low-confidence edges
13	Budget 100% returns the baseline route	PASS	

3. Backend API tests

#	Test	Result
1	GET /health returns ok	PASS
2	GET /sun at 13:00 is about 0.77	PASS

#	Test	Result
3	GET /sun rejects hour 99 with 400	PASS
4	POST /route returns two routes	PASS
5	Coordinates snap to nearest nodes	PASS
6	step_free request works or says NO_ROUTE	PASS
7	Impossible budget returns warning BUDGET_NOT_MET	PASS
8	Night time: no sun, equal routes	PASS
9	Bad node id returns 400	PASS
10	Bad closed edge id returns 400	PASS
11	Missing destination returns 400	PASS
12	All exits closed returns 404 NO_ROUTE	PASS

4. Experiment: 148 random long trips

We picked random start and end points at least 6 blocks apart, at 1 pm, with a 60% heat budget, and compared CoolRoute with the shortest route.

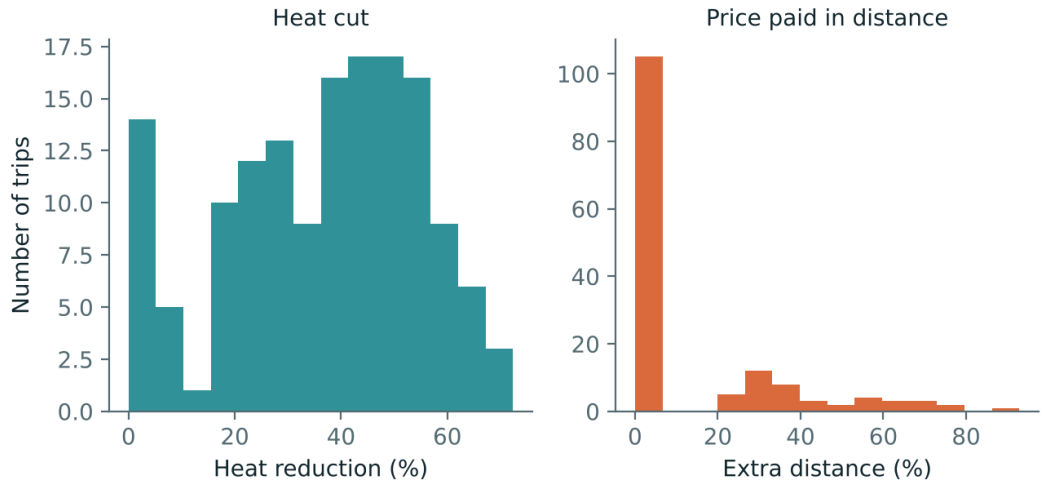


Figure 1. Left: how much heat CoolRoute removed on each trip. Right: how much extra distance it cost.

Measure	Value
Median heat reduction	40%
Mean heat reduction	36%
Median extra distance	3.2%
90th percentile extra distance	43%
Trips where the 60% budget was met	53%
Trips needing more than 20% extra distance	29%

What this tells us (the honest reading)

- **Good:** the typical trip gets about 40% cooler for about 3% more distance. In this synthetic map the gain comes from its two shaded avenues.
- **Limit:** a 60% budget was met on only 53% of trips. On the rest, CoolRoute returns the coolest route it can find and the app says the budget was not met. This is the behaviour the problem statement asks for.

- **Cost:** 29% of trips would need more than 20% extra distance. The app flags these as 'Large detour' so the user can choose.
- **Why:** in our synthetic map only two avenues are well shaded. A real city with more tree cover would meet budgets more often. We have not tested on a real city yet.

5. Heat saved through the day

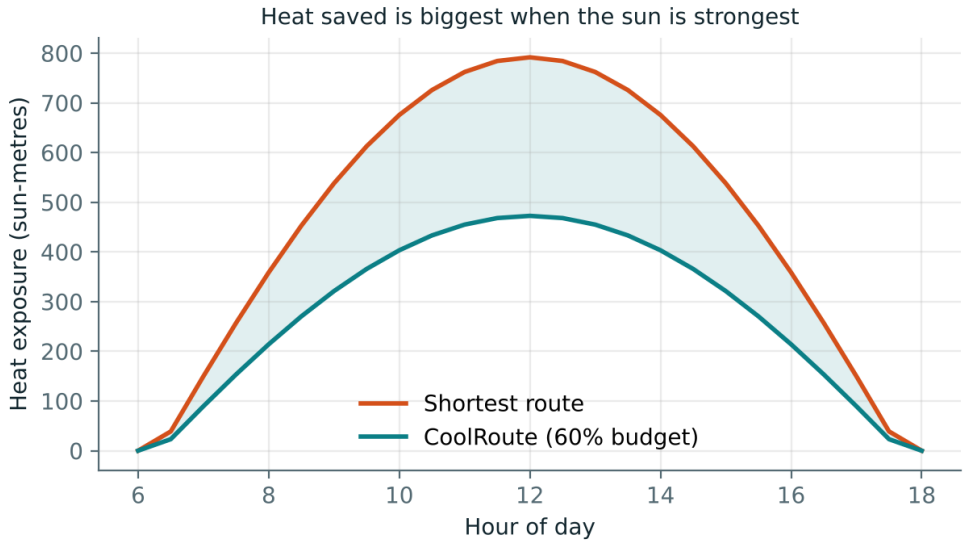


Figure 2. The gap between the lines is the heat CoolRoute saves. It is largest around midday and zero at night.

6. Known limits

- Synthetic neighbourhood and synthetic shade values. Real data will change the numbers.
- The heat score does not include air temperature, humidity or wind. Open-Meteo data can be added.
- Lagrangian search finds a route on the efficient frontier, not always the exact optimum.
- No user testing yet with real pedestrians.
- Shade is fixed per street for an hour. Real shade moves minute by minute.

7. How to repeat these results

```
python3 test_core.py        # 13 engine tests
python3 make_cases.py && node verify.js     # JS vs Python on 60 scenarios
python3 test_api.py         # 12 API tests (start the server first)
python3 make_figs.py        # experiment and charts
```